Q1 Psuedocode

```
FUNCTION solution (A, K)
    IF length of A equals 0
        RETURN empty list

    CREATE rotated list filled with zeros
    SET its length equal to length of A

    FOR each index from 0 to length A-1
        CALCULATE new_index = (index + K) MOD length of A
        SET rotated [new_index] = A [index]
    END FOR

    RETURN rotated
```

Q2 Psuedocode

```
FUNCTION Solution(A)
    CREATE empty dict called counts

    FOR each number in A
        IF number exists in counts
            INCREASE counts[number] by 1

        ELSE
            SET counts[number] = 1

        END IF
    END FOR

    FOR each number in counts
        IF counts[number] MOD 2 equals 1
            RETURN number

        END IF
    END FOR
```

Q3 Psuedocode

```
FUNCTION solution (X, Y, D)
        SET distance = Y - X
        SET jumps = distance DIV D
        If distance MOD D is not equal to 0
            INCREASE jumps by 1

        END IF

        RETURN jumps
```

Q4 Psuedo code

```
FUNCTION Solution (A)
    SET expected_sum = 0
    FOR number from 1 to length of A+1
        ADD number to expected sum

    END FOR

    SET actual_sum = 0
    FOR each number in A
        ADD number to actual_sum

    END FOR

    SET missing = expected_sum - actual_sum
    RETURN missing
```

Q5 Pseudocode

```
FUNCTION solution (A)
    SET n equal to length of A
    CREATE list called seen containing n+1 FALSE values

    FOR each number in A
        IF number is less than 1 or greater than n
            RETURN 0

        END IF

        IF seen[number] is TRUE
            RETURN 0

        END IF

        SET seen[number] = True

    END FOR
```